

Exercise 13: Bayesian Optimization

Optimization in Machine Learning

Reproducibility

Produced with Python 3.10 and R 4.4.1. To reproduce, install the pinned package versions below.

Python

```
pip install numpy==2.0.2 matplotlib==3.10.8 scipy==1.15.3 scikit-learn==1.7.2
```

R

```
install.packages("remotes")
remotes::install_version("ggplot2", "3.5.1")
remotes::install_version("DiceKriging", "1.6.1")
remotes::install_version("patchwork", "1.3.0")
```

Problem 1: Closed-form Expected Improvement

Derive the closed form expression of the **Expected Improvement** acquisition function:

$$a_{\text{EI}}(\mathbf{x}) = (f_{\min} - \hat{f}(\mathbf{x})) \Phi\left(\frac{f_{\min} - \hat{f}(\mathbf{x})}{\hat{s}(\mathbf{x})}\right) + \hat{s}(\mathbf{x}) \phi\left(\frac{f_{\min} - \hat{f}(\mathbf{x})}{\hat{s}(\mathbf{x})}\right).$$

Assume $Y(\mathbf{x}) \sim \mathcal{N}(\hat{f}(\mathbf{x}), \hat{s}^2(\mathbf{x}))$.

Hints:

- Start from $a_{\text{EI}}(\mathbf{x}) = \mathbb{E}_y[\max\{f_{\min} - y, 0\}] = \int \max\{f_{\min} - y, 0\} p(y) dy$.
- Decompose the integral by whether $y < f_{\min}$ or $y \geq f_{\min}$.

- Substitute $u = (y - \hat{f}(\mathbf{x}))/\hat{s}(\mathbf{x})$.
- Use $\int_{-\infty}^z u \phi(u) du = -\phi(z)$.

Problem 2: Implement Bayesian Optimization

Implement BO with a Gaussian-process surrogate and Expected Improvement acquisition.
Goal: minimize

$$f : [0, 1] \rightarrow \mathbb{R}, \quad x \mapsto 2x \sin(14x).$$

Start with 4 points sampled uniformly at random and terminate after 10 total evaluations.

1. Write the BO loop in pseudocode.
2. Implement it. Optimize the EI via a univariate method (Brent's / `optimize`).